

James N. Grossmann III

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EDUCATION

University of Florida

Degrees: B.S. in Physics, B.A. in Political Science

Gainesville, FL

August 2024 – May 2027

- GPA: 4.0/4.0
- Relevant coursework: Quantum Mechanics I, Statistical Mechanics*, Electromagnetism I and II[†], Classical Mechanics I and II, Collider Physics at the Higgs Frontier, Advanced Electronics* and Modern Physics[†] Laboratory, Computational Linear Algebra, Python for Engineers (* = Ongoing, [†] = Planned for Spring 2027)

RESEARCH EXPERIENCE

Undergraduate Researcher in Physics

University of Florida

May 2025 – Ongoing

Experimental Condensed Matter Physics

- Advisor: Prof. James Hamlin
- Performed Monte Carlo searches for a minimized-magnetic susceptibility Cr-Mn-Fe-Co-Ni Cantor alloy composition, ideal for non-magnetic applications
- Synthesized and characterized candidate alloys through arc-melting, quartz tube sealing and annealing, and Superconducting Quantum Interference Device (SQUID) magnetometry, and Magnetic Property Measurement System (MPMS) measurements to compare theoretical and experimental magnetic properties. Awarded the 2026-2027 University of Florida University Scholars Program \$1,750 research grant for the proposal of this project
- Arc-melted, powdered, and performed X-ray diffraction (XRD) preparation and analysis on 50+ samples to evaluate computational database stability predictions, the results of which our group is currently preparing a manuscript for

The Pennsylvania State University Physics REU Student

The Pennsylvania State University

May 2026 – Aug 2026

Experimental Condensed Matter Physics

- Advisor: Prof. Qi Li
- Fabricated micron-scale 2D electron gas (2DEG) Hall bar devices at SrTiO₃ (STO) interfaces through photolithography and sputtering deposition, tracking the time evolution of device mobility and charge carrier density via cryostat measurements
- Investigated the effects of air exposure and sputtering pressure on resulting device mobility, identifying physical parameters that allowed several devices to maintain optimal mobility and carrier density conditions over time
- Trained user of the Penn State Materials Research Institute Nanofabrication Lab (Cleanroom); trained and authorized user of Heidelberg MLA 150 instrument for photolithography processes. Gained experience in cryostat and sputtering operation for characterization and synthesis, respectively

Directed Project in Physics Research

University of Florida

Jan 2026 – May 2026

Experimental High-Energy Physics

- Instructor: Prof. Philip Chang
- Analyzed Large Hadron Collider (LHC) data through Python, C++, and ROOT through the University of Florida HiPerGator supercomputing cluster
- For the capstone project, developed a cutflow and Binary Decision Tree (BDT) classifier machine learning algorithm to isolate the extraordinarily rare di-Higgs signal from Standard Model background in a Compact Muon Solenoid (CMS) dataset
- Best result achieved a 95% confidence level population mean value of $\mu_{95} = 11.535$, demonstrating an effective signal-isolating analysis

POSTERS, PRESENTATIONS, AND PUBLICATIONS

Publications

- G. Di Gianluca, J. Nidel, J. Esquivel, **J.N. Grossmann III**, T. Miniati, M. Crossman, Z. Li, P. Prakash, B. Geisler, J. Kim, R. Hennig, P. Hirschfeld, G. Stewart, J.J. Hamlin, “From Database to Diffractometer: Putting Structure Predictions to the Test.” Manuscript in preparation.
- **J.N. Grossmann III**, “Physics 1 Forever” (self-published textbook), Zenodo.
<https://doi.org/10.5281/zenodo.19667306>

Posters and Presentations

- **J.N. Grossmann III**, A. Heltman, Y. Gao, Q. Li, “Synthesis of Stable High Mobility 2D Electron Gas at SrTiO₃ Interfaces and Electrical Characterization of Hall Bar Devices at Varying Growth Conditions,” 2026 17th Annual Penn State REU Symposium. <https://doi.org/10.5281/zenodo.21519449>
- **J.N. Grossmann III**, J.J. Hamlin, “Searching for a Non-Magnetic Cantor Alloy Composition,” University of Florida Center for Undergraduate Research 2026 Spring Symposium.
- **J.N. Grossmann III**, J.J. Hamlin, “Searching for a Non-Magnetic Cantor Alloy Composition,” University of Florida Department of Physics Research Poster Presentation Session 2026.
- G. Di Gianluca, J. Nidel, J. Esquivel, **J.N. Grossmann III**, T. Miniati, M. Crossman, Z. Li, P. Prakash, B. Geisler, J. Kim, R. Hennig, P. Hirschfeld, G. Stewart, J.J. Hamlin, “From Database to Diffractometer: Putting Structure Predictions to the Test,” American Physical Society (APS) Global Physics Summit 2026. <https://meetings-archive.aps.org/smt/2026/mar-g26/2/>
- G. Di Gianluca, J. Nidel, J. Esquivel, **J.N. Grossmann III**, T. Miniati, M. Crossman, Z. Li, P. Prakash, B. Geisler, J. Kim, R. Hennig, P. Hirschfeld, G. Stewart, J.J. Hamlin, “Synthesizability of Computationally Predicted Novel Materials,” University of Florida Center for Undergraduate Research 2026 Spring Symposium.
- G. Di Gianluca, J. Nidel, J. Esquivel, **J.N. Grossmann III**, T. Miniati, M. Crossman, Z. Li, P. Prakash, B. Geisler, J. Kim, R. Hennig, P. Hirschfeld, G. Stewart, J.J. Hamlin, “Synthesizability of Computationally Predicted Novel Materials,” University of Florida Department of Physics Research Poster Presentation Session 2026.

TEACHING EXPERIENCE

Undergraduate Teaching Assistant in Physics

Jan 2026 – May 2026

University of Florida

Introductory Physics Laboratory Teaching Assistant

- Independently taught ~ 40 students across two sections of Physics 2 laboratory classes in the Spring 2026 semester as a paid position for 10 hours a week, with work split between class instruction, grading, and office hours
- As all teaching work was completely autonomous, gained tremendous experience in face-to-face instructional delivery. Guided students through electromagnetism and optics experiments through emphasis of collaborative problem solving in regards to experimental troubleshooting

LEADERSHIP EXPERIENCE

Student-Led Academic Peer Study

Aug 2024 – Ongoing

University of Florida

Physics Mentor, Director of Senior Mentors, Treasurer

- Mentored students in introductory physics courses for over 200 volunteer hours in small group format, running a student-run support group that served more than 180 physics students. Transformed the direction of the student organization from simple office hour sessions to structured, directed peer-learning sessions targeting problematic areas for students in respective classes, leading to a dramatic increase in attendance of the organization’s sessions
- Managed budgeting, financial planning, and student government funding submissions on behalf of a team of over 20 mentors and their groups of students (2025-2026 academic year)

Society of Physics Students

May 2025 – Ongoing

University of Florida

Vice President, Industry Ambassador

- Promoted academic, research, and industry opportunities to an organization of over 100 physics majors and organized workshops on résumés/CVs, research involvement, and career preparation for physics students. Organized and coordinated undergraduate talks, social events, and mentorship assignments for physics majors at the University of Florida

AWARDS

University Scholars Program: Awarded a \$1,750 research stipend to conduct condensed matter physics research under Prof. James Hamlin during the 2026-2027 academic year (University of Florida, Center for Undergraduate Research)

Gail L. Guth Scholarship: Twice awarded the \$2,000 scholarship for academic merit in the 2025-2026 and 2026-2027 academic years (University of Florida)

Garrett Endowment for Physics Education: Twice awarded the scholarship for academic merit for \$360 and \$330 during the 2025-2026 and 2024-2025 academic years, respectively (University of Florida, Department of Physics)

TECHNICAL SKILLS

Programming: LaTeX, Python (NumPy, SciPy, Matplotlib, pandas, pickle), C++, ROOT

Laboratory: Cleanroom usage, Photolithography (Heidelberg MLA 150), cryostat operation, sputtering, arc-melting, SQUID and MPMS magnetometry, XRD, single-crystal growth, quartz tube sealing and annealing, My Green Lab certified

Software: Adobe (Photoshop, InDesign), kLayout, Office